that gravity couldn't spill into them at all. Under this assumption, gravity appears weak because it *is* weak. But now, if we work in the braneworld scenario and allow the extra dimensions to be much larger than had previously been considered, the observed feebleness of gravity no longer means that it's intrinsically weak. Instead, gravity could be a relatively powerful force that appears weak only because the relatively large extra dimensions, like large pipes, dilute its intrinsic strength. Following this line of reasoning, if gravity is much stronger than once thought, the strings can be much longer than once thought, too.

As of today, the question of exactly how long doesn't have a unique, definite answer. With the newfound freedom to vary both the size of strings and the size of the extra dimensions over a much wider range than previously envisioned, there are a number of possibilities. Dimopoulos and his collaborators have argued that existing experimental results, both from particle physics and from astrophysics, show that unexcited strings can't be larger than about a billionth of a billionth of a meter (10^{-18} meters). While small by everyday standards, this is about a hundred million billion (10^{17}) times larger than the Planck length—nearly a hundred *million billion times larger than previously thought.* As we'll now see, that would be large enough that signs of strings could be detected by the next generation of particle accelerators.

String Theory Confronts Experiment?

The possibility that we are living within a large three-brane is, of course, just that: a possibility. And, within the braneworld scenario, the possibility that the extra dimensions could be much larger than once thought—and the related possibility that strings could also be much larger than once thought—are also just that: possibilities. *But they are tremendously exciting possibilities.* True, even if the braneworld scenario is right, the extra dimensions and the string size could still be Planckian. But the possibility within string/M-theory for strings and the extra dimensions to be much larger—to be just beyond the reach of today's technology—is fantastic. It means that there is at least a chance that in the next few years, string/ M-theory will make contact with observable physics and become an experimental science.